DGC sampling grid on TinyGSM, Sudoku and OWT

CoBit checkpoints, one GH200 per cell, 20-21 Oct 2026, about 30 GPU-h. One sampling seed for GSM8K, three for Sudoku and for the OWT points that decide the front comparison.

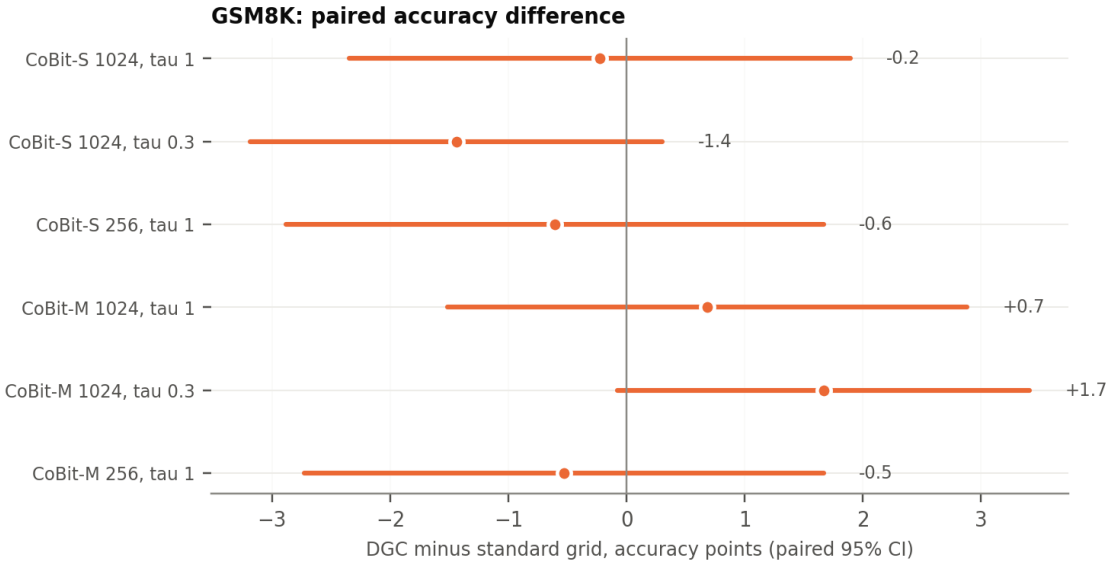
Findings. (1) **OWT: DGC moves the GenPPL-entropy front toward higher entropy.** With gamma swept on both grids in one pipeline (3 seeds at the deciding points), CoBit-S is level at low entropy and DGC is 2-8 GenPPL better above entropy ~5.22 at the same NFE. CoBit-M's two grids never overlap in entropy at a given NFE: allowing any NFE <= 512, the standard grid is better below entropy 5.34 (13.0 vs 14.7), DGC is better above 5.37 (14.7 vs 18.4), and only DGC reaches 5.40 (15.3) or the real-data entropy 5.44 (20.0). (2) **GSM8K: no gain.** At the standard settings every difference is within +/-2 points. DGC prefers less churn (0.3 instead of 0.41), but its best cell (35.2%) ties the standard grid's best (35.3%). (3) **Sudoku: one real gain**, hard puzzles with churn, 66.1% -> 70.0% (+3.9 [+2.8, +5.0]). (4) **Mechanism: two effects pulling opposite ways.** On Sudoku the gain comes from where the churn acts (DGC's churn on the standard steps: +2.2; DGC's steps with the standard churn: -0.2). On OWT the extra entropy comes from where the steps are (DGC's steps with the standard churn give the highest entropy), while DGC's low-sigma churn lowers both entropy and GenPPL.

What changed and what didn't. Only the sigma grid: the DGC step density (Wainwright, arXiv:2607.26285), rebuilt from each checkpoint's own entropy-rate table by scripts/dgc/build_dgc_tables.py, replaces the entropy-rate grid. Checkpoint, sampler, churn gamma, NFE, seeds and metrics are those of each task's existing protocol. Because the churn window is read from the same table, per-step gamma is unchanged but its placement follows the grid. DGC moves steps from sigma > 5 to sigma 0.2-1 and puts more steps below 0.17 (TinyGSM 1024 steps: 7 -> 47 below 0.17, 209 -> 52 above 5; last real step 0.157 -> 0.093).

TinyGSM -> GSM8K (1319 problems, seed 0, EMA)

checkpoint	setting	standard %	DGC %	DGC - std (95% CI)	invalid tok % (std / DGC)
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 1	28.1	27.9	-0.2 [-2.4, +1.9]	0.001 / 0.001
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 0.3, unsharpened last step	35.3	33.8	-1.4 [-3.2, +0.3]	0.000 / 0.000
CoBit-S unguided 500k	256 steps, churn 0.41, tau 1	26.8	26.2	-0.6 [-2.9, +1.7]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.41, tau 1	33.3	34.0	+0.7 [-1.5, +2.9]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.41, tau 0.3, unsharpened last step	38.1	39.7	+1.7 [-0.1, +3.4]	0.000 / 0.000
CoBit-M 250k	256 steps, churn 0.41, tau 1	31.5	30.9	-0.5 [-2.7, +1.7]	0.001 / 0.001

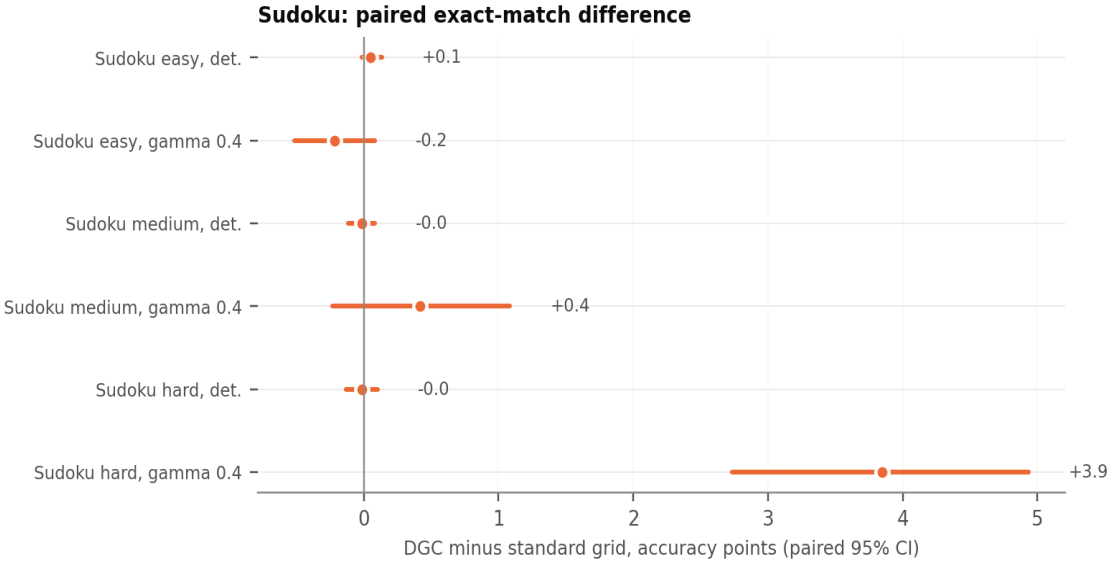
CoBit-S 1024-step standard-grid cells are the stored score-temperature cells (results/scoretemp); the gate reran the first 128 problems and matched every answer. CoBit-M's stored cells used the Karras grid, so both of its arms were run here.



Sudoku (2000 validation puzzles, 180 steps, raw weights, sigma_data 0.5)

difficulty	sampler	standard % (seeds 42-44)	DGC % (seeds 42-44)	DGC - std (95% CI)	n
easy	deterministic	93.4 (93.5, 93.9, 93.0)	93.5 (93.5, 94.0, 93.0)	+0.1 [-0.0, +0.1]	3
easy	churn gamma 0.4	98.5 (98.4, 98.4, 98.6)	98.2 (98.1, 98.0, 98.6)	-0.2 [-0.5, +0.1]	3
medium	deterministic	76.8 (76.4, 77.5, 76.5)	76.8 (76.5, 77.5, 76.3)	-0.0 [-0.1, +0.1]	3
medium	churn gamma 0.4	91.4 (91.5, 91.2, 91.6)	91.9 (91.8, 91.7, 92.0)	+0.4 [-0.2, +1.1]	3
hard	deterministic	40.4 (40.9, 39.1, 41.2)	40.4 (40.9, 39.2, 41.2)	-0.0 [-0.1, +0.1]	3
hard	churn gamma 0.4	66.1 (65.6, 65.2, 67.5)	70.0 (70.8, 70.0, 69.2)	+3.9 [+2.7, +4.9]	3

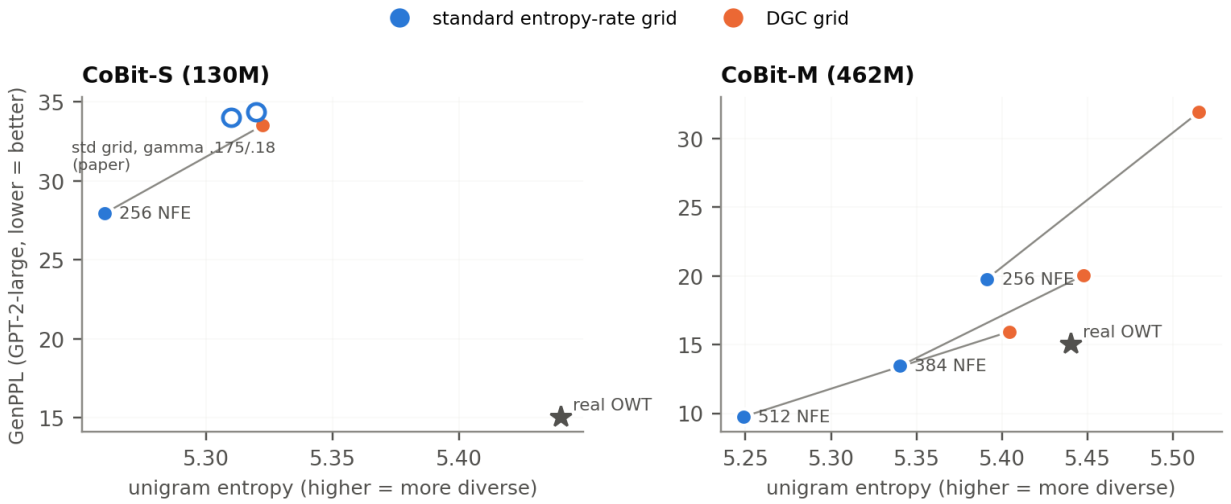
Raw weights because the 20k-step EMA scores 0%; gamma 0.4 is where the stored gamma sweep plateaus. CI pools the three seeds' paired puzzles.



OWT at the paper's gamma (1024 samples, seed 42, terminal sigma 0.08)

model	NFE / gamma	paper GenPPL / H	standard GenPPL / H	DGC GenPPL / H	dGenPPL	dH
CoBit-S	256 / 0.13	27.06 / 5.26	27.95 / 5.260	33.50 / 5.322	+5.55	+0.062
CoBit-M	256 / 0.21	19.48 / 5.40	19.78 / 5.391	31.91 / 5.515	+12.13	+0.124
CoBit-M	384 / 0.24	13.06 / 5.33	13.44 / 5.340	20.03 / 5.447	+6.59	+0.107
CoBit-M	512 / 0.26	9.87 / 5.25	9.75 / 5.249	15.94 / 5.404	+6.19	+0.155

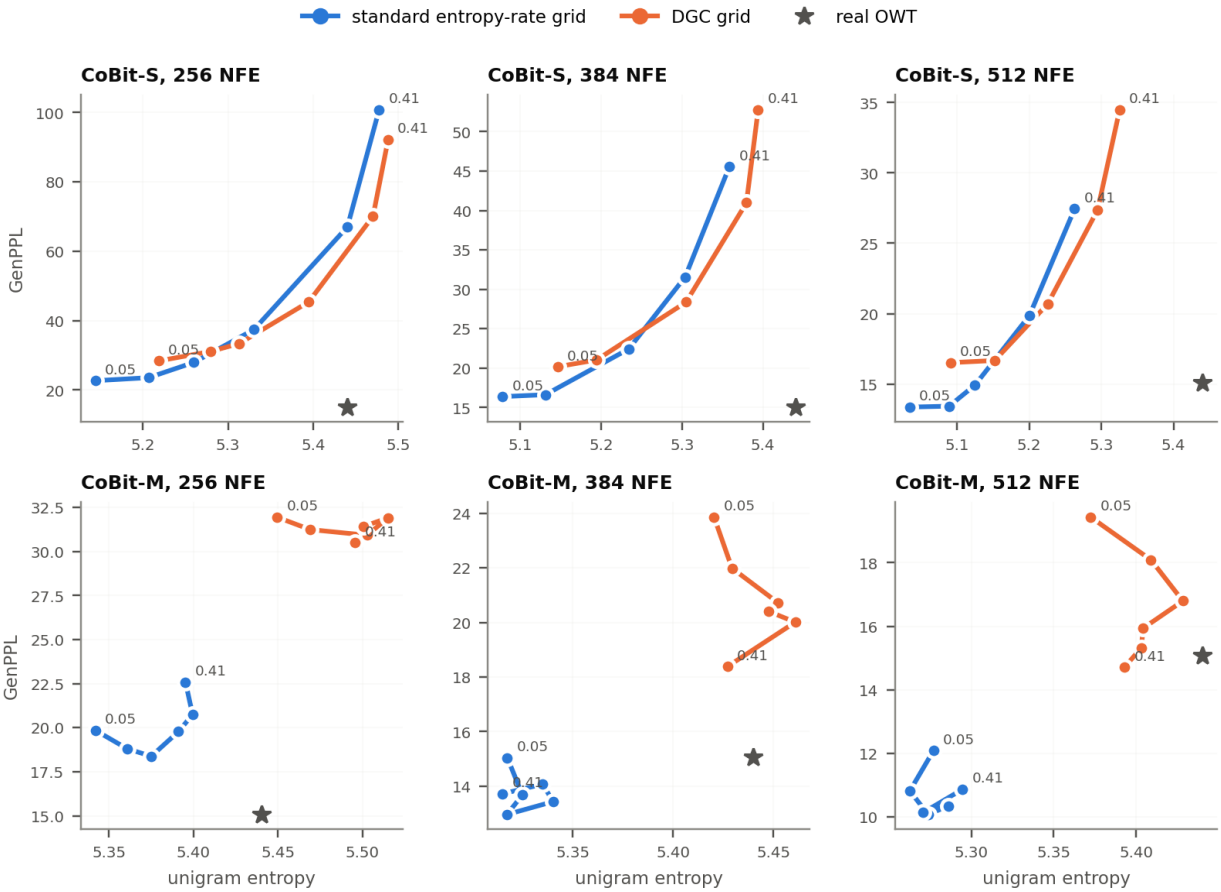
Same pipeline for both arms, without the real-data pass (no OWT cache on Isambard); real OWT reference 15.07 / 5.44 from the paper. GenPPL is a corpus-level number from one seed: the paper's seed-to-seed sd is about 0.6 (CoBit-S, 256 NFE). Lower GenPPL at lower entropy is a move along the frontier, not an improvement, so read both columns. The mechanism section shows where DGC's extra entropy comes from: its step placement, not its churn placement.



Each grey segment joins the two grids at the same NFE and gamma. Hollow blue points: the standard grid at higher churn, from the paper (CoBit-S only).

OWT Pareto fronts: gamma swept on both grids

Both grids swept over gamma {0.05, 0.1, 0.2, 0.3, 0.41} at NFE 256, 384 and 512 (plus the paper's gamma above), same pipeline, 1024 samples, seed 42 plus seeds 43-44 at the 24 points that decide the comparison. The comparison that matters is GenPPL at equal entropy: each grid's sweep is interpolated linearly in entropy and compared at three entropies spanning the range both grids reach.



Each line is one grid's gamma sweep at fixed NFE, joined in gamma order; the end points are labelled with gamma.

model	NFE	entropy	standard GenPPL	DGC GenPPL	DGC - std
CoBit-S	256	5.284	31.10	31.37	+0.27
CoBit-S	256	5.348	42.19	38.52	-3.67
CoBit-S	256	5.413	59.63	51.28	-8.35
CoBit-S	384	5.199	20.47	21.37	+0.90
CoBit-S	384	5.252	24.83	24.89	+0.06
CoBit-S	384	5.306	31.99	28.46	-3.53
CoBit-S	512	5.134	15.52	16.62	+1.10
CoBit-S	512	5.177	18.34	18.03	-0.31
CoBit-S	512	5.220	22.24	20.38	-1.86
CoBit-M	256	no overlap (5.449 > 5.399)	-	-	-
CoBit-M	384	no overlap (5.420 > 5.340)	-	-	-
CoBit-M	512	no overlap (5.372 > 5.295)	-	-	-

Points with seeds 43-44 are 3-seed means; the rest are seed 42. Neighbouring points move by about 1-2 GenPPL from seed noise alone.

Best front over NFE <= 512

At fixed NFE the CoBit-M curves of the two grids do not overlap in entropy (gamma barely moves CoBit-M's entropy), so the fair comparison is the best each grid can do at a required entropy, allowing any swept NFE and gamma.

model	entropy	standard: GenPPL (NFE, gamma)	DGC: GenPPL (NFE, gamma)
CoBit-S	>= 5.20	19.9 (512, 0.3)	20.7 (512, 0.2)
CoBit-S	>= 5.25	27.4 (512, 0.41)	27.4 (512, 0.3)
CoBit-S	>= 5.30	31.6 (384, 0.3)	28.4 (384, 0.2)
CoBit-S	>= 5.35	45.6 (384, 0.41)	41.0 (384, 0.3)
CoBit-S	>= 5.40	67.0 (256, 0.3)	70.1 (256, 0.3)
CoBit-S	>= 5.44	67.0 (256, 0.3)	70.1 (256, 0.3)
CoBit-M	>= 5.30	13.0 (384, 0.2)	14.7 (512, 0.41)
CoBit-M	>= 5.34	13.4 (384, 0.24)	14.7 (512, 0.41)
CoBit-M	>= 5.37	18.4 (256, 0.2)	14.7 (512, 0.41)
CoBit-M	>= 5.40	45.2 (128, 0.2)	15.3 (512, 0.3)
CoBit-M	>= 5.44	45.2 (128, 0.2)	20.0 (384, 0.3)

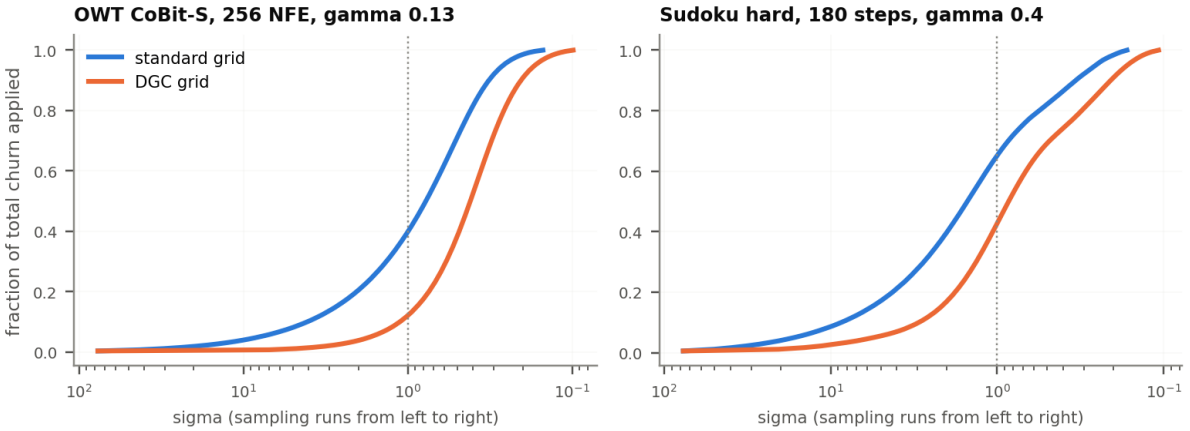
TinyGSM: DGC with re-tuned churn and temperature (CoBit-S unguided 500k)

checkpoint	setting	standard %	DGC %	DGC - std (95% CI)	invalid tok % (std / DGC)
CoBit-S unguided 500k	1024 steps, churn 0.2, tau 0.3, unsharpened last step	33.4	34.0	+0.7 [-1.1, +2.5]	0.000 / 0.000
CoBit-S unguided 500k	1024 steps, churn 0.3, tau 0.3, unsharpened last step	33.0	35.2	+2.2 [+0.4, +4.0]	0.000 / 0.000
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 0.2, unsharpened last step	35.1	33.3	-1.8 [-3.7, +0.1]	0.000 / 0.000
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 0.5, unsharpened last step	34.6	33.4	-1.1 [-3.0, +0.8]	0.000 / 0.000
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 0.7, unsharpened last step	32.4	31.5	-0.9 [-2.9, +1.1]	0.000 / 0.000

Standard-grid partners are the stored score-temperature cells (blocks A and C), except tau 0.7, which was run here on both grids.

Mechanism: where the steps are vs where the churn acts

With full-band churn every step gets the same gamma, so the churn per unit log-sigma follows the step density: moving the steps moves the churn. The swapped arms separate the two. The steps stay on one table's grid while gamma_i = gamma * p_other(sigma_i) / p_grid(sigma_i) (capped at sqrt(2)-1), so the churn density follows the other table (COBIT_CHURN_TABLE in DDIMSampler).



Cumulative churn along the trajectory. The dotted line is sigma = 1: the DGC grid applies most of its churn below it, the standard grid much less.

steps	churn placement	hard Sudoku % (seeds 42-44)	vs standard/standard (95% CI)
standard	standard	62.0 (62.3, 61.5, 62.2)	-
standard	DGC	64.1 (64.7, 63.2, 64.5)	+2.2 [+1.6, +2.8]
DGC	standard	61.8 (61.8, 61.1, 62.5)	-0.2 [-1.4, +0.9]
DGC	DGC	64.9 (65.2, 64.5, 65.0)	+2.9 [+1.8, +4.0]

Hard Sudoku, churn gamma 0.15, chosen so the cap barely binds in the swapped arms.

gamma	steps	churn placement	GenPPL	entropy
0.05	standard	standard	22.68	5.145
0.05	standard	DGC	20.23	5.102
0.05	DGC	standard	35.08	5.272
0.05	DGC	DGC	28.38	5.219
0.13	standard	standard	27.95	5.260
0.13	standard	DGC	19.07	5.132
0.13	DGC	standard	50.91	5.405
0.13	DGC	DGC	33.50	5.322

OWT CoBit-S, 256 NFE, seed 42, 1024 samples.

Part 2: everything at T = 1

Added 21 Oct (evening), 8.3 GPU-h (39.3 GPU-h for the whole study). No score temperature anywhere; each grid gets its own best churn, so the comparisons are best-vs-best. Everything above is unchanged.

> **Part 2 findings (T = 1, best churn per grid).** (1) **TinyGSM: no gain.** CoBit-S: the standard grid's best is 29.0% (churn 0.3), DGC's best 27.9% (churn 0.41); DGC is equal or lower at every churn. CoBit-M: 33.8% (standard, churn 0.3) vs 34.0% (DGC, churn 0.3 and 0.41), +0.2. Only one paired difference excludes zero (CoBit-S, churn 0.3: -2.4). (2) **Sudoku: the hard-puzzle gain holds best-vs-best**: 70.3% vs 66.2% (+4.1, both at gamma 0.5), with DGC ahead at every gamma from 0.1 to 0.5; medium +0.4, easy -0.2. (3) **LM1B, both grids swept in one pipeline: the OWT pattern again.** At 128 NFE DGC is far better at matched entropy (about 82 vs 94-107 GenPPL at entropy 4.34). At 256 and 512 NFE the two grids do not overlap in entropy: DGC sits 0.02-0.05 higher, the standard grid reaches lower GenPPL (512 NFE: 43.6 at 4.24 vs 46.4 at 4.295). Across all NFE <= 512, DGC is ahead at every entropy from 4.28 up: 48.3 vs 59.5 GenPPL at entropy >= 4.30, and 61.0 vs 86.8 near the real-data entropy (>= 4.34); the standard grid's lowest GenPPL comes only at lower entropy. Our DGC points reproduce Georgios's sweep (512 NFE: 46.4 / 4.295 vs 46.3 / 4.291), and our standard 256-NFE point matches the paper (59.6 vs 59.76), not his 57.3.

At T = 1: TinyGSM with churn re-tuned for each grid

checkpoint	setting	standard %	DGC %	DGC - std (95% CI)	invalid tok % (std / DGC)
CoBit-S unguided 500k	1024 steps, churn 0.1, tau 1	26.8	26.4	-0.5 [-2.7, +1.7]	0.001 / 0.001
CoBit-S unguided 500k	1024 steps, churn 0.2, tau 1	28.0	27.6	-0.4 [-2.4, +1.7]	0.001 / 0.002
CoBit-S unguided 500k	1024 steps, churn 0.3, tau 1	29.0	26.6	-2.4 [-4.5, -0.3]	0.001 / 0.000
CoBit-S unguided 500k	1024 steps, churn 0.41, tau 1	28.1	27.9	-0.2 [-2.4, +1.8]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.1, tau 1	28.8	30.6	+1.8 [-0.5, +4.2]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.2, tau 1	32.0	31.0	-1.0 [-3.1, +1.2]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.3, tau 1	33.8	34.0	+0.2 [-1.9, +2.4]	0.001 / 0.001
CoBit-M 250k	1024 steps, churn 0.41, tau 1	33.3	34.0	+0.7 [-1.5, +2.9]	0.001 / 0.001

No score temperature. Best-vs-best means each grid's best churn row, not the same row.

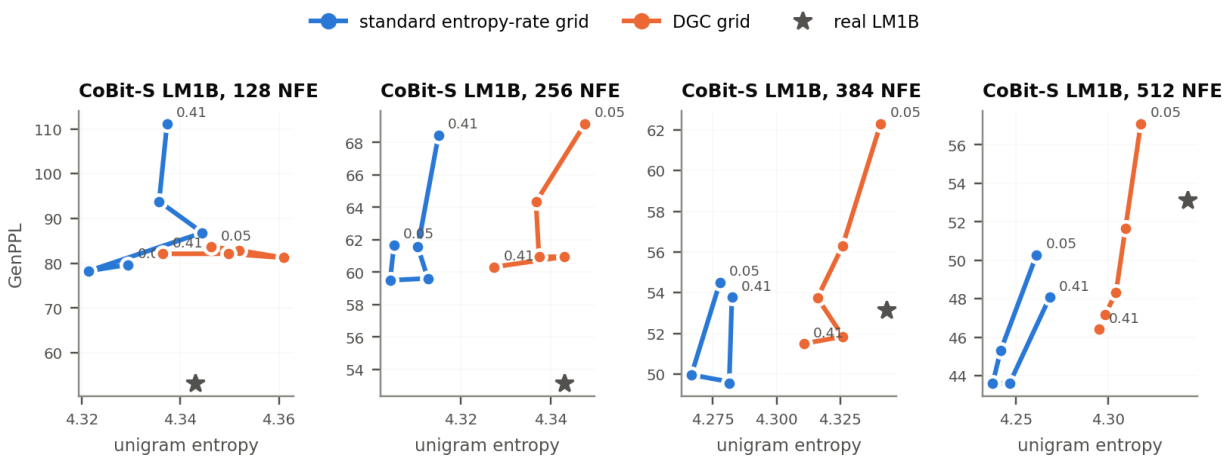
At T = 1: Sudoku with gamma swept on both grids

difficulty	grid	det.	0.1	0.2	0.3	0.4	0.5	best
easy	standard	93.4	97.4	98.1	98.4	98.5	98.4	98.5 (gamma 0.4)
easy	DGC	93.5	96.6	97.6	97.9	98.2	98.3	98.3 (gamma 0.5)
medium	standard	76.8	88.0	90.1	91.1	91.4	91.5	91.5 (gamma 0.5)
medium	DGC	76.8	87.6	90.0	91.0	91.9	91.8	91.9 (gamma 0.4)
hard	standard	40.4	59.4	63.2	65.1	66.1	66.2	66.2 (gamma 0.5)
hard	DGC	40.4	61.5	66.8	68.1	70.0	70.3	70.3 (gamma 0.5)

Exact-match %, mean of seeds 42-44, 2000 puzzles, 180 steps, raw weights.

LM1B Pareto fronts: gamma swept on both grids, this pipeline

Released CoBit-S LM1B checkpoint, NFE 128-512 x gamma {0.05, 0.1, 0.2, 0.3, 0.41}, both grids, seed 42, 1024 samples, GH200. Unlike the LM1B sweep in results/dgc/lm1b_sweeps (DGC only, compared with the paper's points from another pipeline), both grids here come from one pipeline. The real-data marker is the same-code measurement from that sweep (53.13 / 4.343); this cluster has no LM1B cache.



Each line is one grid's gamma sweep at fixed NFE, joined in gamma order.

model	NFE	entropy	standard GenPPL	DGC GenPPL	DGC - std
CoBit-S LM1B	128	4.338	107.24	82.41	-24.84
CoBit-S LM1B	128	4.340	100.43	82.72	-17.71
CoBit-S LM1B	128	4.342	93.63	83.04	-10.59
CoBit-S LM1B	256	no overlap (4.327 > 4.315)	-	-	-
CoBit-S LM1B	384	no overlap (4.311 > 4.282)	-	-	-
CoBit-S LM1B	512	no overlap (4.295 > 4.268)	-	-	-

model	entropy	standard: GenPPL (NFE, gamma)	DGC: GenPPL (NFE, gamma)
CoBit-S LM1B	>= 4.28	49.5 (384, 0.3)	46.4 (512, 0.41)
CoBit-S LM1B	>= 4.30	59.5 (256, 0.1)	48.3 (512, 0.2)
CoBit-S LM1B	>= 4.32	78.3 (128, 0.1)	51.8 (384, 0.3)
CoBit-S LM1B	>= 4.34	86.8 (128, 0.2)	61.0 (256, 0.3)

One seed (42). Second table: lowest GenPPL among all swept points with at least that entropy.

Part 3: finite-NFE DP-DGC (CoBit-M OWT)

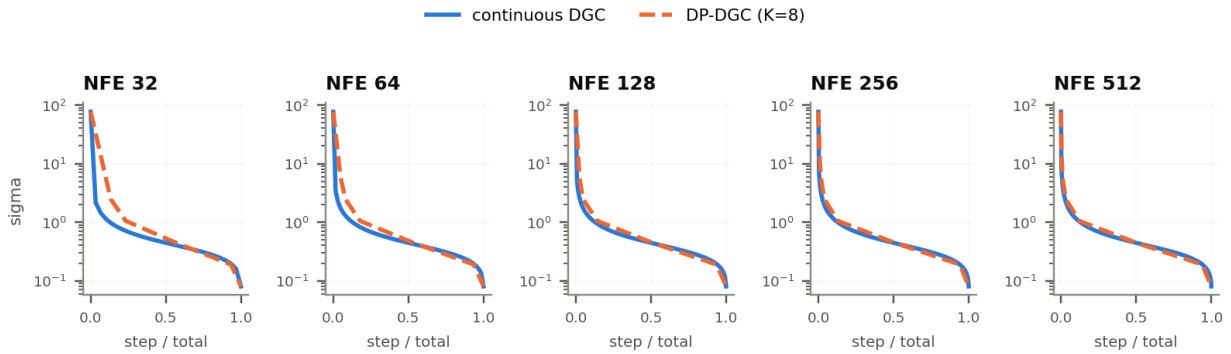
Added 22 Oct. Question: does solving the finite-NFE DGC allocation problem with a DP beat the continuous DGC grid at low NFE? Only the sigma grid changes: same CoBit-M checkpoint, DGC table, endpoints (76.76 to 0.08), DDIM sampler, churn gamma 0.2 at every NFE, T = 1, seed 42, 1024 samples.

Part 3 findings: no evidence that the finite-NFE DP helps. (1) **Low NFE:** at NFE 32 DP-DGC is worse than continuous DGC (GenPPL 385.9 vs 361.7 at the same entropy); at 64 and 128 it is 5.0 and 1.8 GenPPL lower but also 0.011 and 0.007 lower in entropy, within one-seed noise at GenPPL 70-170. (2) **High NFE:** DP-DGC is 0.5-1.1 GenPPL lower and 0.016-0.043 lower in entropy, a move along the front: at 256, 384 and 512 NFE a continuous-DGC point from the Part 1 gamma sweep beats it on both axes at once (e.g. 512 NFE: 14.9 / 5.399 at gamma 0.3 vs DP 16.2 / 5.385). (3) **Why so little changes:** with K = 8 the exponential term is nearly linear in the blocks that hold the steps (S_k / n_k ~ 0.17 even at NFE 32), and the near-empty high-sigma blocks have G_k ~ 0, so the DP solution is the continuous allocation plus one forced step in each high-sigma block and log-uniform steps within blocks. The finite-NFE problem as posed here is not where the low-NFE losses come from. One seed, gamma fixed at 0.2, S_k and G_k derived without the reference notebook.

The DP grid

K = 8 blocks uniform in log sigma^2. DP minimises sum_k G_k (exp(S_k / n_k) - 1) over integers n_k >= 1 with sum n_k = N - 1 (the sampler's interval count), with S_k the block width in log sigma^2 and G_k = m_k^2 / S_k, m_k the DGC table mass in the block, so the large-N optimum is the continuous DGC allocation (n_k proportional to m_k). Each block is then filled with n_k log-uniform steps. The reference notebook (DGC_five_delta_schedulers.ipynb) was not available; S_k and G_k are this derivation, not the notebook's.

NFE	intervals	DP allocation (high sigma first)	continuous share x intervals
32	30	1, 1, 1, 1, 3, 10, 11, 2	0.0, 0.0, 0.2, 0.6, 2.7, 11.4, 13.3, 1.8
64	62	1, 1, 1, 2, 6, 22, 25, 4	0.0, 0.0, 0.3, 1.3, 5.6, 23.6, 27.5, 3.7
128	126	1, 1, 1, 3, 12, 46, 54, 8	0.0, 0.1, 0.7, 2.6, 11.3, 47.9, 56.0, 7.5
256	254	1, 1, 2, 6, 23, 95, 111, 15	0.1, 0.1, 1.3, 5.2, 22.8, 96.5, 112.8, 15.1
384	382	1, 1, 3, 9, 35, 143, 167, 23	0.1, 0.2, 2.0, 7.8, 34.3, 145.2, 169.7, 22.7
512	510	1, 1, 3, 11, 46, 192, 225, 31	0.1, 0.2, 2.7, 10.5, 45.8, 193.9, 226.6, 30.3



The DP grid keeps >= 1 step in each of the near-empty high-sigma blocks, where the continuous grid jumps from 76 to about 2 in one step, and is log-uniform within blocks.

Model metric: GenPPL / entropy

NFE	standard	continuous DGC	DP-DGC (K=8)	DP - continuous
32	472.7 / 5.670	361.7 / 5.677	385.9 / 5.681	+24.2 / +0.004
64	139.5 / 5.570	169.7 / 5.632	164.7 / 5.621	-5.0 / -0.011
128	45.2 / 5.483	72.5 / 5.571	70.7 / 5.564	-1.8 / -0.007
256	18.4 / 5.375	31.1 / 5.503	30.0 / 5.487	-1.1 / -0.016
384	12.8 / 5.308	20.7 / 5.452	20.2 / 5.423	-0.5 / -0.030
512	10.1 / 5.274	16.8 / 5.428	16.2 / 5.385	-0.6 / -0.043

GenPPL (GPT-2-large) / unigram entropy, seed 42 for every cell, so the three grids share initial noise. Standard and continuous-DGC cells at 256-512 NFE are the seed-42 cells of the Part 1 sweep. One seed: about 1 GenPPL is within noise.

Part 4: DP-DGC with the reference notebook's G_k

Added 22 Oct, after DGC_five_delta_schedulers.ipynb became available. Part 3's G_k was a derivation (m_k^2 / S_k); the notebook defines G_k = 1/2 int_block h'(v) dr, the block's DGC increment. With the table's density p(u) ∝ sqrt(h'), this is G_k ∝ int_block p(u)^2 du. Blocks, S_k, the DP and the within-block geometric steps were already the notebook's. Same protocol as Part 3.

Part 4 findings (run 23 Oct): the notebook's G_k does not change Part 3's conclusion. Its allocations are within a few steps of Part 3's (identical at NFE 32) and so are the samples: 385.9 at NFE 32 (continuous DGC 361.7, worse), and from NFE 64 to 512 between 0.9 and 4.5 GenPPL below continuous DGC but always at lower entropy (-0.012 to -0.056), i.e. a move along the front, not past it. The DP's 15x lower Theorem-1 bound at NFE 32 again does not show up in GenPPL.

NFE	intervals	DP allocation, notebook G_k (high sigma first)	Theorem-1 bound, continuous	Theorem-1 bound, DP
32	30	1, 1, 1, 1, 3, 10, 11, 2	0.00996	0.000678
64	62	1, 1, 1, 2, 6, 23, 24, 4	0.00121	0.000286
128	126	1, 1, 1, 3, 13, 48, 51, 8	0.000233	0.000133
256	254	1, 1, 2, 6, 25, 98, 105, 16	7.25e-05	6.4e-05
384	382	1, 1, 3, 9, 38, 148, 158, 24	4.23e-05	4.22e-05
512	510	1, 1, 4, 12, 50, 198, 212, 32	3e-05	3.15e-05

Theorem-1 bound (the notebook's finite-step KL surrogate, sum over intervals of ((hi/lo)^2 - 1) G) on each grid, up to the arbitrary scale of the table; the DP minimises it over blockwise-geometric grids. It is a surrogate, not the model metric.

NFE	standard	continuous DGC	DP (Part 3 G_k)	DP (notebook G_k)	notebook DP - continuous
32	472.7 / 5.670	361.7 / 5.677	385.9 / 5.681	385.9 / 5.681	+24.2 / +0.004
64	139.5 / 5.570	169.7 / 5.632	164.7 / 5.621	167.3 / 5.621	-2.4 / -0.012
128	45.2 / 5.483	72.5 / 5.571	70.7 / 5.564	68.0 / 5.556	-4.5 / -0.016
256	18.4 / 5.375	31.1 / 5.503	30.0 / 5.487	29.0 / 5.480	-2.1 / -0.024
384	12.8 / 5.308	20.7 / 5.452	20.2 / 5.423	19.8 / 5.420	-0.9 / -0.032
512	10.1 / 5.274	16.8 / 5.428	16.2 / 5.385	15.6 / 5.372	-1.2 / -0.056

GenPPL / unigram entropy, seed 42, churn gamma 0.2, T = 1, 1024 samples.

Part 5: pass@1 at low NFE on the tasks

Added 22 Oct. Can the finite-NFE DP grid (K = 8, the notebook's G_k) raise single-sample accuracy at low step counts? Standard vs continuous DGC vs DP-DGC, T = 1. On these tables the grid runs to sigma 0.002 and the DGC density is ~0 below sigma ~0.1, so the DP's n_k >= 1 puts single steps at sigma ~0.03 and ~0.008 where the continuous grid jumps from ~0.1 to 0.002.

Part 5 findings: the finite-NFE DP grid does not raise pass@1; on Sudoku it lowers it. (1) **GSM8K:** at 64, 128 and 256 steps, on CoBit-S and CoBit-M, every DP-DGC difference is within +/-1.4 points and every 95% CI contains zero (vs continuous DGC and vs standard). The three grids are within noise of each other at these step counts. (2) **Sudoku hard:** continuous DGC beats the standard grid at every step count (+2.3 to +4.1). DP-DGC loses 1.0-2.4 points of that against continuous DGC at every step count, each CI excluding zero, and sits at the standard grid's level at 32-64 steps. (3) **Likely reason, untested:** Part 1's mechanism test put the Sudoku gain in DGC's low-sigma churn placement. The DP keeps the block totals but spaces steps log-uniformly within blocks and spends two forced steps at sigma ~0.03 and ~0.008, below where the table has density, so it changes exactly the low-sigma part the gain comes from. Invalid-token rates are unchanged (<= 0.002%).

GSM8K (1319 problems, seed 0, churn 0.41, EMA)

model	steps	standard % (inv. tok %)	continuous DGC %	DP-DGC %	DP - continuous	DP - standard
CoBit-S	64	20.2 (0.001)	20.0 (0.002)	20.9 (0.001)	+0.9 [-1.0, +2.7]	+0.8 [-1.4, +2.9]
CoBit-S	128	24.0 (0.001)	24.0 (0.001)	23.8 (0.001)	-0.2 [-2.3, +1.8]	-0.2 [-2.4, +2.0]
CoBit-S	256	26.8 (0.001)	26.2 (0.001)	25.9 (0.001)	-0.4 [-2.5, +1.7]	-1.0 [-3.3, +1.3]
CoBit-M	64	24.2 (0.002)	24.0 (0.001)	22.8 (0.001)	-1.1 [-3.2, +0.8]	-1.4 [-3.6, +0.9]
CoBit-M	128	27.8 (0.001)	28.6 (0.001)	28.1 (0.001)	-0.5 [-2.4, +1.6]	+0.3 [-1.9, +2.4]
CoBit-M	256	31.5 (0.001)	30.9 (0.001)	30.6 (0.001)	-0.3 [-2.4, +1.8]	-0.8 [-3.1, +1.4]

Accuracy is pass@1 (one sample per problem); paired 95% CIs over problems. Invalid-token rate in brackets. CoBit-S 256-step standard and continuous cells are Part 1's.

Sudoku hard (2000 puzzles, gamma 0.5, raw weights, seeds 42-44)

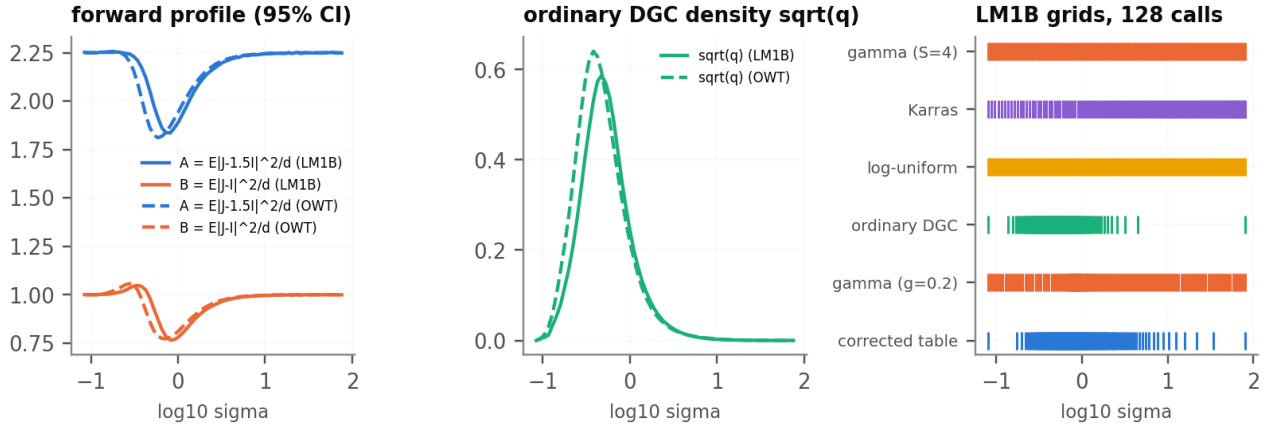
steps	standard %	continuous DGC %	DP-DGC %	DP - continuous	DP - standard
32	52.7	55.0	52.6	-2.4 [-3.0, -1.8]	-0.1 [-1.1, +1.0]
64	57.9	60.9	58.7	-2.2 [-2.9, -1.6]	+0.7 [-0.5, +1.9]
90	61.4	63.9	62.9	-1.0 [-1.5, -0.4]	+1.5 [+0.4, +2.7]
180	66.2	70.3	69.3	-1.1 [-1.6, -0.5]	+3.1 [+2.0, +4.2]

Exact match, mean of three seeds; paired CIs pool the seeds' puzzles. The 180-step standard and continuous cells are Part 2's.

Part 6: gamma-dependent Euler grid (CoBit-S LM1B and OWT)

Added 23 Oct, from the handoff euler_dgc_derivation.pdf. Grid density per unit log sigma p = sqrt(A / (S^2 B + eta)), S = N gamma, with A = E||J - 1.5 I||^2 / d and B = E||J - I||^2 / d measured forward-only on held-out data (2048 sequences, one Brownian path each, 64 log-sigma bins on [0.08, 80], EMA weights, null self-conditioning). Every arm is a precomputed grid with exact endpoints 0.08 and 80, inverted with the cumulative mass at bin edges. NFE is the number of model calls actually made (64/128/256 calls = 63/127/255 intervals; the legacy NFE label counts one more). 1024 samples, seeds 42-46 paired across arms; cells show seed-mean GenPPL / entropy and, in brackets, the paired difference from the corrected table with a t 95% interval over seeds.

Part 6 findings: the gamma-dependent grid does not help; the profile-based ordinary DGC grid does, at low NFE. (1) **The profile is nearly flat.** On both checkpoints A ~ 2.25 and B ~ 1 (a negligible Jacobian) except for a dip at sigma ~0.3-1, so sqrt(A / (S^2 B + eta)) is close to log-uniform at S = 1-12 and only concentrates on the dip at 256 calls with production churn (S = 33-51). (2) **Production sampler (DDIM + EDM churn).** The gamma grid ties log-uniform at 64 and 128 calls, both far behind the corrected table (LM1B +65 / +35 GenPPL; OWT +242-262 / +123-138). At 256 calls it is level on LM1B (-2.0 +/- 0.7 GenPPL at gamma 0.185, but entropy -0.025 to -0.039) and worse on OWT (+7 to +12, entropy not higher). (3) **Ordinary DGC from the same profile, density sqrt(q), is the best grid at low NFE.** 64 calls: LM1B -20 / -21 GenPPL, OWT -23 / -67 / -78 at the three gammas, entropy equal or higher. 128 calls: LM1B -7.7 at +0.02 entropy; OWT level (+1.7 to -5.3, CIs ~ +/-3) at +0.03-0.04 entropy. 256 calls: LM1B level at +0.03 entropy; OWT +5 GenPPL at +0.05-0.06 entropy, i.e. the Part 1 shift of the front toward higher entropy (at matched entropy ~5.30, DGC gamma 0.13 gives 31.7 vs the table's 33.2 at gamma 0.175). (4) **Literal Euler (theory sweep) is far from production quality** (LM1B 230-480, OWT 500-1100 GenPPL; null self-conditioning, small churn). Under it the corrected table is best, gamma ~ log-uniform, Karras worse, and the DGC grid collapses (one 3.2-nat step from sigma 80): the grid ranking under the surrogate's own solver does not predict the ranking under the production sampler. Sanity check: the corrected table at 256 calls, gamma 0.13, gives OWT 26.5 / 5.241 against the paper's 27.06 / 5.26.



Held-out data: LM1B test split through the training pipeline (detokenise, bert-base-uncased, [SEP] boundaries, 128-token blocks, 15-bit codes); OWT train[-100000:] (the documents held out from training) through GPT-2 ids, the bpe16 code tokenizer and 16-bit codes. The K = 128 profile (same sequences, fresh path), averaged onto the K = 64 bins, agrees to within 4% for A and B and 20% for q, but about half its bins fall outside the K = 64 sequence-bootstrap CIs: the bin-width dependence of the finite-bin estimator is visible, though small next to the grid differences.

LM1B: theory sweep, literal Euler at fixed S

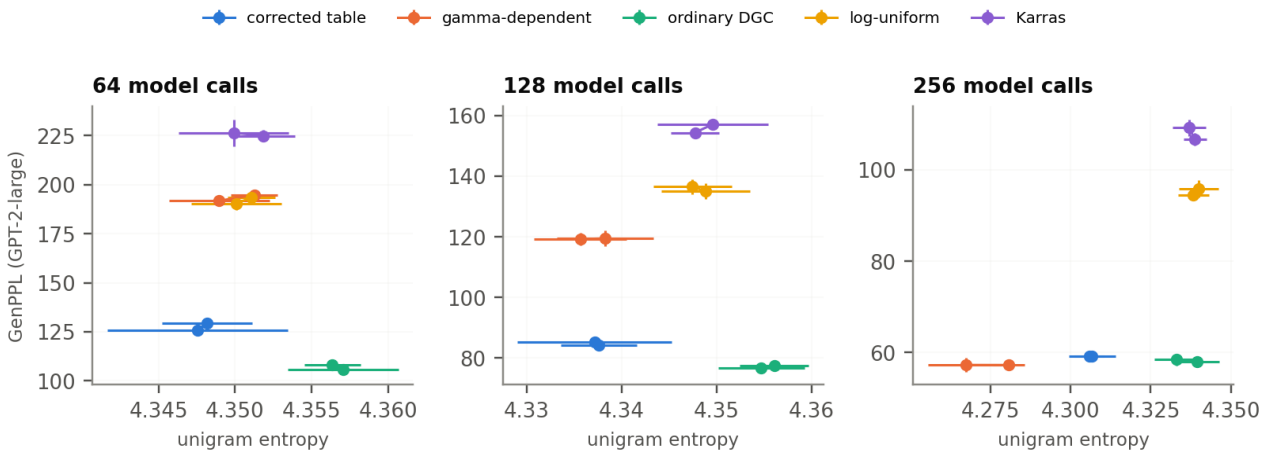
calls	grid	S = 1 (gamma = 1/N)	S = 4 (gamma = 4/N)
64	corrected table	346.3 / 4.392	322.8 / 4.377
	gamma-dependent	370.3 / 4.386 (+24.1±4.0 / -0.006)	443.2 / 4.410 (+120.4±6.0 / +0.033)
	ordinary DGC	5387.5 / 4.725 (+5041.2±53.7 / +0.333)	4599.3 / 4.609 (+4276.5±63.7 / +0.232)
	log-uniform	365.9 / 4.385 (+19.7±3.7 / -0.007)	436.6 / 4.408 (+113.8±5.1 / +0.031)
	Karras	388.2 / 4.386 (+42.0±6.2 / -0.006)	483.4 / 4.417 (+160.6±4.9 / +0.040)
128	corrected table	274.7 / 4.361	260.9 / 4.359
	gamma-dependent	299.5 / 4.364 (+24.8±4.8 / +0.003)	315.8 / 4.374 (+54.9±4.6 / +0.015)
	ordinary DGC	4133.6 / 4.640 (+3858.9±62.5 / +0.280)	3765.3 / 4.613 (+3504.4±56.3 / +0.254)
	log-uniform	297.3 / 4.363 (+22.6±4.5 / +0.002)	311.7 / 4.373 (+50.8±4.9 / +0.014)
	Karras	307.8 / 4.363 (+33.1±1.9 / +0.003)	328.5 / 4.375 (+67.6±4.4 / +0.016)
256	corrected table	249.9 / 4.348	232.7 / 4.346
	gamma-dependent	268.6 / 4.351 (+18.7±5.6 / +0.003)	264.4 / 4.357 (+31.7±5.5 / +0.011)
	ordinary DGC	691.9 / 4.488 (+442.0±9.6 / +0.140)	1017.0 / 4.490 (+784.3±15.1 / +0.144)
	log-uniform	266.9 / 4.350 (+17.0±4.6 / +0.002)	261.0 / 4.355 (+28.3±4.8 / +0.009)
	Karras	274.1 / 4.353 (+24.2±4.2 / +0.004)	272.6 / 4.355 (+40.0±7.1 / +0.009)

x_next = x + (1 + lambda) sigma Delta (D - x) / sigma^2 + sqrt(2 lambda sigma Delta) z, lambda_i = gamma sigma_i / Delta_i; the denoiser is evaluated before noise is added; null self-conditioning (the closure the profile was measured with).

LM1B: production sampler (DDIM + EDM churn, carry self-conditioning)

calls	grid	gamma = 0.185	gamma = 0.2
64	corrected table	125.7 / 4.348	129.4 / 4.348
	gamma-dependent	191.8 / 4.349 (+66.1±3.3 / +0.001)	194.9 / 4.351 (+65.5±4.0 / +0.003)
	ordinary DGC	105.6 / 4.357 (-20.0±2.7 / +0.009)	107.9 / 4.356 (-21.4±1.3 / +0.008)
	log-uniform	190.5 / 4.350 (+64.8±4.5 / +0.003)	193.4 / 4.351 (+64.0±2.3 / +0.003)
	Karras	224.9 / 4.352 (+99.2±2.6 / +0.004)	226.5 / 4.350 (+97.1±7.0 / +0.002)
128	corrected table	84.3 / 4.338	85.2 / 4.337
	gamma-dependent	119.5 / 4.338 (+35.2±2.3 / +0.001)	119.3 / 4.336 (+34.1±3.3 / -0.001)
	ordinary DGC	76.6 / 4.355 (-7.7±1.4 / +0.017)	77.5 / 4.356 (-7.7±2.2 / +0.019)
	log-uniform	135.0 / 4.349 (+50.7±2.4 / +0.011)	136.5 / 4.347 (+51.3±3.7 / +0.010)
	Karras	154.4 / 4.348 (+70.1±2.1 / +0.010)	157.1 / 4.350 (+71.9±2.1 / +0.012)
256	corrected table	59.2 / 4.306	59.1 / 4.307
	gamma-dependent	57.3 / 4.281 (-2.0±0.7 / -0.025)	57.3 / 4.267 (-1.9±1.5 / -0.039)
	ordinary DGC	58.4 / 4.333 (-0.8±1.1 / +0.027)	57.9 / 4.340 (-1.2±1.3 / +0.033)
	log-uniform	94.4 / 4.338 (+35.2±0.8 / +0.033)	95.9 / 4.340 (+36.7±2.9 / +0.033)
	Karras	106.7 / 4.339 (+47.4±1.4 / +0.033)	109.2 / 4.337 (+50.1±1.6 / +0.030)

The EDM transfer: each grid replaces only the sigma points; churn gamma acts at every step (s_churn = gamma x intervals). The gamma-dependent grid uses S = N x gamma of its column.



OWT: theory sweep, literal Euler at fixed S

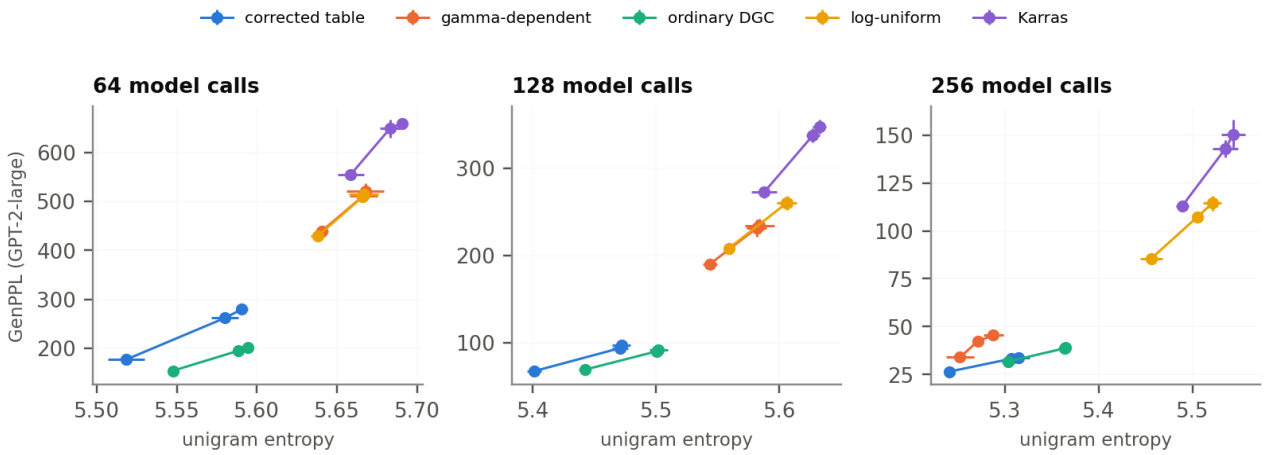
calls	grid	S = 1 (gamma = 1/N)	S = 4 (gamma = 4/N)
64	corrected table	806.6 / 5.636	996.7 / 5.666
	gamma-dependent	825.1 / 5.682 (+18.5±22.7 / +0.047)	984.3 / 5.739 (-12.4±18.0 / +0.073)
	ordinary DGC	33081.2 / 6.682 (+32274.6±235.2 / +1.046)	53861.9 / 6.378 (+52865.2±604.5 / +0.712)
	log-uniform	811.8 / 5.679 (+5.2±20.6 / +0.044)	963.9 / 5.733 (-32.8±17.7 / +0.068)
	Karras	894.7 / 5.692 (+88.1±18.0 / +0.056)	1103.1 / 5.756 (+106.4±27.2 / +0.090)
128	corrected table	580.2 / 5.601	609.6 / 5.629
	gamma-dependent	639.1 / 5.624 (+58.9±19.3 / +0.023)	657.4 / 5.654 (+47.8±24.2 / +0.025)
	ordinary DGC	38926.6 / 6.266 (+38346.5±515.0 / +0.664)	18028.8 / 6.023 (+17419.2±219.7 / +0.394)
	log-uniform	637.5 / 5.623 (+57.3±18.4 / +0.022)	655.5 / 5.653 (+45.9±25.2 / +0.025)
	Karras	664.2 / 5.630 (+84.1±13.9 / +0.028)	716.8 / 5.669 (+107.2±17.1 / +0.040)
256	corrected table	522.2 / 5.586	496.1 / 5.602
	gamma-dependent	557.4 / 5.595 (+35.2±8.9 / +0.009)	540.8 / 5.610 (+44.7±10.7 / +0.008)
	ordinary DGC	1190.0 / 5.570 (+667.8±51.9 / -0.016)	4634.0 / 5.338 (+4137.9±152.4 / -0.264)
	log-uniform	555.6 / 5.594 (+33.4±9.3 / +0.009)	537.7 / 5.611 (+41.5±11.7 / +0.010)
	Karras	573.9 / 5.593 (+51.8±14.5 / +0.008)	561.7 / 5.616 (+65.5±16.6 / +0.015)

x_next = x + (1 + lambda) sigma Delta (D - x) / sigma^2 + sqrt(2 lambda sigma Delta) z, lambda_i = gamma sigma_i / Delta_i; the denoiser is evaluated before noise is added; null self-conditioning (the closure the profile was measured with).

OWT: production sampler (DDIM + EDM churn, carry self-conditioning)

calls	grid	gamma = 0.13	gamma = 0.175	gamma = 0.18
64	corrected table	176.9 / 5.518	261.6 / 5.580	279.5 / 5.591
	gamma-dependent	439.0 / 5.640 (+262.1±9.9 / +0.122)	511.6 / 5.667 (+250.0±7.5 / +0.087)	521.5 / 5.668 (+242.0±14.4 / +0.077)
	ordinary DGC	154.0 / 5.548 (-22.9±5.5 / +0.029)	194.6 / 5.588 (-67.0±8.8 / +0.008)	201.0 / 5.594 (-78.5±9.7 / +0.004)
	log-uniform	429.4 / 5.638 (+252.5±10.0 / +0.120)	509.4 / 5.666 (+247.8±5.9 / +0.086)	513.9 / 5.667 (+234.4±14.2 / +0.076)
	Karras	554.9 / 5.659 (+378.0±11.9 / +0.140)	649.0 / 5.683 (+387.4±19.4 / +0.103)	659.1 / 5.691 (+379.7±12.5 / +0.100)
128	corrected table	67.2 / 5.401	93.7 / 5.471	96.6 / 5.472
	gamma-dependent	190.1 / 5.544 (+122.9±3.9 / +0.143)	231.4 / 5.582 (+137.6±12.5 / +0.111)	233.8 / 5.584 (+137.1±8.2 / +0.112)
	ordinary DGC	68.9 / 5.442 (+1.7±1.8 / +0.041)	90.0 / 5.500 (-3.7±3.1 / +0.030)	91.4 / 5.502 (-5.3±3.6 / +0.030)
	log-uniform	207.9 / 5.559 (+140.7±5.3 / +0.158)	260.1 / 5.606 (+166.3±5.6 / +0.135)	260.2 / 5.606 (+163.6±8.3 / +0.134)
	Karras	272.5 / 5.588 (+205.3±7.4 / +0.187)	337.6 / 5.627 (+243.9±11.4 / +0.156)	348.0 / 5.633 (+251.4±9.6 / +0.161)
256	corrected table	26.5 / 5.241	33.2 / 5.307	33.6 / 5.314
	gamma-dependent	33.9 / 5.252 (+7.4±1.5 / +0.011)	42.4 / 5.271 (+9.2±0.9 / -0.036)	45.5 / 5.288 (+11.8±1.3 / -0.027)
	ordinary DGC	31.7 / 5.303 (+5.2±0.7 / +0.062)	38.7 / 5.364 (+5.4±1.2 / +0.057)	39.2 / 5.365 (+5.6±0.5 / +0.051)
	log-uniform	85.6 / 5.456 (+59.0±2.3 / +0.215)	107.4 / 5.505 (+74.2±2.8 / +0.198)	114.4 / 5.521 (+80.8±4.1 / +0.207)
	Karras	113.1 / 5.489 (+86.6±2.8 / +0.248)	143.1 / 5.535 (+109.9±5.3 / +0.228)	150.4 / 5.543 (+116.8±7.6 / +0.229)

The EDM transfer: each grid replaces only the sigma points; churn gamma acts at every step (s_churn = gamma x intervals). The gamma-dependent grid uses S = N x gamma of its column.



Part 7: TinyGSM with the profile-based grids

Added 23 Oct. Part 6's grids on the prompted task: forward profile on 2048 held-out TinyGSM validation sequences (prompt clean and clamped into the self-condition, answer and padding noised, EMA, the eval's sigma_data), 64 log-sigma bins on [0.002, 80], the eval's full band. Production sampler (DDIM + EDM churn 0.41, carry, T = 1), seed 0, all 1319 GSM8K problems. The gamma-dependent grid uses S = (steps - 1) x 0.41. Steps = model calls with the carry closure.

Part 7 findings: on GSM8K neither profile-based grid raises pass@1; the gamma grid lowers it. (1) **The TinyGSM profile is flat.** A = 2.24-2.25, B = 0.997-1.001 on both checkpoints: about three quarters of the generated positions are padding the model predicts exactly from the prompt, so the per-coordinate average is dominated by J ~ 0 (max q 0.005, vs ~0.36 on LM1B). The gamma grid is then log-uniform on [0.002, 80] apart from the churn term, and it loses at every step count on both models: -1.9 to -7.2 points, 7 of 8 CIs exclude zero. (2) **Ordinary DGC from the profile ties the corrected table** (within +/-1.4 points in 7 of 8 cells; CoBit-S 256 steps -2.7 [-4.9, -0.5]); its best cell, CoBit-M 1024 steps, 35.4% vs 34.0% (+1.4 [-0.8, +3.6]). Unlike LM1B and OWT there is no low-NFE gain: sqrt(q) is shape-only, and its shape is close to the entropy-rate table's. (3) The corrected (edge-paired) table matches the legacy grid within ~1.3 points. (4) **CoBit-M tau 0.3 second seed:** 39.0% vs seed 0's 38.1% (+1.0 [-0.7, +2.7]); mean 38.6%, so tau 0.3's ~5-point gain over tau 1 (33.3%) is not a sampling-seed artefact. Possible follow-up (not run): profile the answer tokens only (exclude padding from gen_mask) for the gamma grid.

model	steps	legacy grid %	corrected table %	ordinary DGC %	gamma grid %	DGC - table	gamma - table
CoBit-S 500k	64	20.2	20.1	19.6	16.2	-0.5 [-2.7, +1.7]	-3.9 [-5.9, -1.9]
	128	24.0	23.7	23.8	19.0	+0.1 [-2.1, +2.2]	-4.8 [-6.8, -2.7]
	256	26.8	27.6	24.9	21.9	-2.7 [-4.9, -0.5]	-5.7 [-7.8, -3.6]
	1024	28.1	27.9	29.1	26.0	+1.2 [-1.1, +3.5]	-1.9 [-3.9, +0.2]
CoBit-M 250k	64	24.2	23.8	24.7	17.7	+0.9 [-1.4, +3.3]	-6.1 [-8.2, -4.0]
	128	27.8	28.5	28.1	21.3	-0.4 [-2.7, +1.9]	-7.2 [-9.5, -4.9]
	256	31.5	30.2	29.9	27.1	-0.3 [-2.6, +2.0]	-3.1 [-5.5, -0.8]
	1024	33.3	34.0	35.4	31.0	+1.4 [-0.8, +3.6]	-3.0 [-5.1, -0.8]

Legacy grid: the earlier standard entropic-grid cells (Parts 1, 2 and 5; midpoint-paired CDF, one fewer model call). Differences are paired over problems, bootstrap 95% CI.

CoBit-M 250k: second seed at tau 0.3 (1024 steps, churn 0.41, standard grid)

setting	accuracy %	95% CI / paired difference
tau 1 (seed 0)	33.3	[30.7, 35.9]
tau 0.3, unsharpened last step, seed 0	38.1	[35.5, 40.6]
tau 0.3, unsharpened last step, seed 1	39.0	[36.5, 41.7]
tau 0.3, mean of seeds 0 and 1	38.6	seed 1 - seed 0: +1.0 [-0.7, +2.7]

Seed 1 repeats the Part 1 cell (score temperature tau 0.3, global mode, unsharpened last step) with a different sampling seed, to separate the tau effect from sampling noise.

Provenance

- Launcher scripts/dgc/tasks_ab.sh (phases gate, main-tasks, main-owt, sweep-owt, seeds-owt, tgsm-screen, mechanism); cell scripts scripts/hpc/arch/{scoretemp_eval,dgc_sudoku,dgc_owt}.slurm.
- Swapped churn: COBIT_CHURN_TABLE in diffusion/continuous/samplers.py (DDIMSampler only; off unless set). LM1B sweep from Georgios: results/dgc/lm1b_sweeps/.

- Part 2: phases t1, lm1b-gate and lm1b-sweep; LM1B config configs/lm1b/continuous/eval/dgc_sweep.py and launcher scripts/hpc/arch/dgc_lm1b.slurm. Jobs died at start-up on faulty compute nodes (bus error or I/O error loading torch/libstdc++ on nid010003, nid010029, nid010033, nid010038, nid010040) and were resubmitted with those nodes excluded (DGC_EXCLUDE); others were held at launch and released.
- Part 3: schedule dp_dgc (diffusion/continuous/dp_dgc.py; tests/test_dp_dgc.py), OWT config SCHEDULE=dp_dgc, phases dp-smoke and dp-owt; schedules dumped by scripts/dgc/dp_dgc_schedules.py.
- Part 5: phase dp-tasks (grid arm dpdgc in gsm8k and dgc_sudoku launchers; SU_STEPS for Sudoku).
- Part 6: held-out data scripts/dgc/build_{lm1b,owt}_heldout_bits.py; profile scripts/dgc/profile_forward_increments.py (diffusion/continuous/forward_profile.py); grids scripts/dgc/build_gamma_grid.py (diffusion/continuous/gamma_grid.py); runs configs/gamma_euler_eval.py via scripts/hpc/arch/gamma_euler_eval.slurm and manifests from scripts/dgc/gamma_euler_plan.sh; samplers euler_lambda / ddim_grid; tests/test_gamma_euler.py. Outputs in results/dgc/gamma_euler/.
- Part 7: held-out data scripts/dgc/build_tinygsm_heldout_bits.py; profile with --task; grids and cells scripts/dgc/tinygsm_ge.sh (gsm8k_eval --grid_file); results results/dgc/gsm8k/*__ge_*.json.
- Code was uncommitted at submission; diff and table hashes in logs/arch/dgc_provenance/.
- DGC tables: assets/entropy_tables/{owt,owt_medium}_dgc and assets/entropy_tables/tasks/*_dgc.